

JAWAHARLAL NEHRU UNIVERSITY
SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

Minor

M.C.A. (Monsoon Semester-2023)

Course Name: Computer Architecture

Maximum Marks: 30

Course Code: CS 405

Total Time: 1 hours

Q1. a) What is the difference between Von Neumann Architecture and Harvard Architecture?. What are the different types of addressing modes that has evolved in Computer Systems?.

b) A 32 bit floating point binary number has 4 bit plus sign for exponent and 26 plus sign for mantissa. The mantissa is a normalized function. Numbers in mantissa and exponent are in a signed-magnitude representation. What are the largest and smallest positive quantities that can be represented excluding zero?.

(10)

Q2. a) What is the difference between RISC and CISC type of Computer Architecture?. Explain with examples.

b) Which one is better and Why?. How can one compensate the drawbacks of one over the other type of architecture.

(10)

Q3. a) What is T- state, Machine Cycle and Instruction Cycle in a Computer System?.

b) Explain the concept of pipelining technique in a computer Architecture system?.

(10)

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JAWAHARLAL NEHRU UNIVERSITY
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Minor Exam

M.CA. (Winter Semester-2024)

Course Name: Computer Architecture

Maximum Marks: 30

Course Code: CS 106

Total Time: 1 hour

Q1. What is the difference between RISC and CISC type of Computer Architecture?.

A computing task is divided into three parts:

1. Part A: 50% of the task can be fully parallelized.
2. Part B: 30% of the task can be parallelized.
3. Part C: 20% of the task is inherently serial and cannot be parallelized.

Calculate the overall speedup when using 8 processors for Part A and 4 processors for Part B. Part C remains serial.

(4+6)

Q2. (a) What are the major performance parameters that determines the execution time of processor?.

b) What is Harvard Architecture?.

- (a) Suppose a program is running on a machine with the following instruction types, CPI values and the frequencies of occurrence. The CPU designer gives two options i) reduce CPI instruction type A to 1.0 and ii) reduce CPI instruction type B to 1.5. ii) reduce CPI instruction type C to 1.8. Which one is better?.

Type	CPI	Frequency
A	1.3	40%
B	2.3	40%
C	2.1	20%

(5+2+3)

Q3. (a) What are the different types of addressing modes that in 8086 microprocessor?.

- (b) Write an assembly language program to subtract two numbers using 2^{nd} complement. What is the new memory capacity if the RAM memory is scaled upto from 2^{20} byte to 2^{28} byte?.

(4+4+2)

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Minor Exam

M.CA. (Winter Semester-2025)

Course Name: Computer Architecture

Maximum Marks: 25

Course Code: CS 405

Total Time: 50 minutes

- Q1. a) What is the difference between RISC and CISC type of Computer Architecture?
b) What is the role of different buses in a Computer system. Explain using 8085 microprocessor.

(3+2)

- Q2. (a) What are the major performance parameters that determines the execution time of processor?
b) Explain which performance parameters effects CPI and why?
(a) Suppose a program is running on a machine with the following instruction types, CPI values and the frequencies of occurrence. The CPU designer gives two options i) reduce CPI instruction type A to 1.0 and ii) reduce CPI instruction type B to 1.5. ii) scale upCPI instruction type C to 2.4. Which one is better?

Type	CPI	Frequency
A	1.4	30%
B	2.5	40%
C	2.1	30%

(4+2+4)

- Q3. (a) What are the different types of machine cycle in 8085 microprocessor?

- (b) Write an assembly language program to subtract two 4 bit numbers. Determine the total T-state and calculate the total time consumed using 8085 microporocessor. What is the new memory capacity if the RAM memory is scaled upto from 2^{20} byte to 2^{34} byte?.

(4+4+2)

JAWAHARLAL NEHRU UNIVERSITY

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Minor Exam

M.CA. (Winter Semester-2026)

Course Name: Computer Architecture

Maximum Marks: 30

Course Code: CS 405

Total Time: 50 minutes

- Q1. a) What is the difference between RISC and CISC type of Computer Architecture?.
- b) What are different addressing modes in 8086 microprocessor. What improvements one sees in 8086 microprocessor compared to 8085 microprocessor?.

(4+6)

Q2. (a) What are the different machine cycle in 8085?.

b) Explain which performance parameters effects CPI , IC and why?.

c) Suppose a program is running on a machine with the following instruction types, CPI values and the frequencies of occurrence. The CPU designer gives two options i) reduce CPI instruction type A to 1.0 and ii) reduce CPI instruction type B to 1.5. ii) scale up CPI instruction type C to 2.4. Which one is better?.

Type	CPI	Frequency
A	1.7	20%
B	2.2	40%
C	2.1	40%

(4+2+4)

Q3. (a) A processor has **5 pipeline stages**, and each stage takes **25 ns** to complete. Assume there is no pipeline overhead.

A program has to process **100 independent instructions**.

Calculate:

1. The total execution time **without pipelining**.
2. The total execution time **with pipelining**.
3. The **speedup** obtained using pipelining.
4. The **pipeline efficiency**.
5. The **throughput** of the pipelined processor.

b) Explain the working of 4 bit pipeline ripple carry adder.

(6+4)